

Notes: Nadauld & Weisbach (JFE, 2012)
Did securitization affect the cost of corporate debt?

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1 Big picture

- **Question.** Did the rise of corporate loan securitization through collateralized loan obligations (CLOs) lower the cost of corporate debt in the primary syndicated-loan market?
- **Main setting.** The paper studies U.S. syndicated corporate loan facilities, especially leveraged loans and Term Loan B facilities, during the 1998–2007 period, with particular attention to the 2002–2007 securitization boom.
- **Core hypothesis.** CLOs created demand for loans that were useful as collateral. If banks could originate loans that CLOs wanted to buy, then this secondary-market demand should raise banks’ willingness to supply those loans in the primary market and lower spreads paid by borrowers.
- **Main empirical result.** Loan facilities later identified as securitized are associated with roughly 17 basis point lower all-in-drawn spreads than otherwise similar non-securitized facilities.
- **Securitization-friendly attributes.** The paper identifies three features that make facilities more likely to be securitized: Term Loan B structure, B-rated borrower debt, and origination by a securitization-active bank.
- **Mechanism test.** Spreads fall especially for B-rated Term Loan B facilities originated by securitization-active banks, exactly where CLO collateral demand should be strongest.
- **Within-loan test.** The paper compares Term Loan B spreads with Term Loan A or revolver spreads within the same loan deal. Because these facilities share the same borrower and many deal-level risks, this comparison is closer to isolating collateral-demand effects.
- **Economic interpretation.** Securitization can affect real corporate financing costs, not only by shifting risk after origination, but by changing the terms at which debt is created in the primary market.
- **Policy relevance.** The findings imply that securitization booms can expand and cheapen credit for the types of debt that are easiest to repackage, creating both potential financing benefits and possible incentives to originate loans for distribution rather than retention.

2 Incremental contribution of the paper

- **Moves securitization research beyond mortgages.** Earlier crisis-era work focused heavily on mortgage securitization. This paper studies corporate bank loans and CLOs, showing that securitization also matters in corporate credit markets.
- **Links secondary-market collateral demand to primary-market loan pricing.** The key contribution is not merely showing that CLOs hold particular loans. It tests whether demand from CLO investors feeds back into the loan-origination market and lowers spreads for borrowers.
- **Builds a novel loan-CLO match.** The authors match Dealscan syndicated loan facilities to Moody’s CLO collateral data, allowing them to identify which corporate loan facilities appear in CLO collateral pools.
- **Identifies securitization-friendly loan attributes.** The paper translates institutional knowledge of CLO collateral preferences into empirical proxies: Term Loan B payoff structure, B-rated credit quality, and securitization-active arrangers.
- **Uses within-deal facility comparisons.** By comparing Term Loan B facilities to Term Loan A facilities or revolvers inside the same loan package, the paper reduces borrower-level confounding and tests whether the discount is specific to the facility type valued by CLOs.
- **Distinguishes CLO-demand from bank risk-tolerance stories.** If securitization-active banks simply became more risk tolerant, discounts should also appear in other facilities they originate. Instead, the evidence is concentrated in securitization-friendly Term Loan B facilities.

- **Documents economic magnitude.** The roughly 20–30 basis point discounts are large for syndicated loan markets. For a \$400 million Term Loan B, 20.9 basis points corresponds to close to \$1 million in annual interest savings.
- **Connects structured finance to the cost of capital.** The paper provides evidence that securitization can lower firms’ borrowing costs through asset-demand pressure, thereby affecting corporate financing decisions and not just intermediary balance sheets.

3 Data and measurement

3.1 Loan-facility data

- The primary loan data come from Dealscan and cover syndicated loan facilities originated in the United States.
- The sample excludes financial firms with SIC codes 6000–6500.
- The paper focuses on loan facilities with Moody’s senior debt ratings Ba1, Ba2, Ba3, B1, B2, and B3.
- Second-lien and mezzanine facilities are removed to focus on comparable senior facilities.
- The main pricing outcome is the **all-in-drawn spread** over LIBOR.

Interpretation: Dealscan gives primary-market contract terms. The unit of observation is a facility, not necessarily an entire loan deal. This matters because the same loan package can contain a revolver, a Term Loan A, and a Term Loan B.

3.2 CLO collateral data

- The authors use a Moody’s database containing the collateral securities in Moody’s-rated CLOs as of the first quarter of 2009.
- A loan facility is classified as securitized if it appears as collateral in the Moody’s CLO database.
- The CLO database is a snapshot, so it may miss facilities securitized earlier and later removed from CLOs.
- This measurement issue likely understates the true amount of securitization and attenuates direct estimates of securitization effects.

Interpretation: Matching loans to CLO holdings lets the authors observe actual securitization, but the snapshot nature of the data motivates the paper’s use of securitization-friendly attributes as an additional identification strategy.

3.3 Facility types

- **Term Loan B** facilities are bullet-like institutional tranches and are especially attractive to CLOs.
- **Term Loan A** facilities are generally amortizing and more bank-held.
- **Revolvers** contain borrower drawdown options and shorter expected maturities, making them less attractive as CLO collateral.

Interpretation: The facility type is central because CLOs need predictable collateral cash flows. Term Loan Bs better fit that need than amortizing Term Loan As or discretionary revolvers.

3.4 Securitization-active banks

- Banks are classified as securitization-active if any lead arranger is among the top ten CLO underwriters.
- The idea is that banks that underwrite CLOs know which loan characteristics are valuable in CLO structures and can originate loans with resale into CLOs in mind.
- Table 2 shows that CLO underwriting is highly concentrated among a small set of banks.

Interpretation: This variable proxies for a bank’s ability and incentive to originate loans for securitization. It is not perfect, but it is grounded in market structure.

3.5 Credit rating classification

- The paper separates B-rated senior debt from Ba-rated senior debt.
- B-rated loans are lower credit quality and are more frequently held by CLOs.
- CLOs may particularly value B-rated collateral because structured finance can transform pools of lower-rated claims into highly rated tranches.

Interpretation: Rating categories help test whether the spread discount appears where CLO demand should be strongest.

3.6 Controls

- Loan and borrower controls include term structure, credit spread, covenant indicator, pricing grid indicator, syndicate size, tranche maturity, log facility amount, log borrower sales, and secured indicator.
- Fixed effects include tranche-purpose fixed effects, year fixed effects, industry fixed effects, credit-rating fixed effects, and in some specifications lead-arranger fixed effects.
- Standard errors are clustered by borrower and year in the key spread regressions.

Interpretation: The controls address observable differences in credit risk, loan structure, macro credit conditions, and loan purpose. They cannot eliminate all unobserved heterogeneity, which motivates the within-loan tests.

4 Models and empirical specifications

4.1 Direct securitization-pricing regression

$$Spread_{i,t} = \alpha + \beta Securitized_{i,t} + \delta' X_{i,t} + FE + \varepsilon_{i,t}. \quad (1)$$

- $Spread_{i,t}$ is the all-in-drawn spread of facility i at origination.
- $Securitized_{i,t}$ equals one if the facility later appears in the Moody’s CLO collateral database.
- $X_{i,t}$ includes loan, borrower, and market controls.

Interpretation: The coefficient β asks whether facilities that ultimately enter CLOs have lower spreads. This is suggestive but not fully causal because securitization may be correlated with unobservable loan demand or risk.

4.2 Probit model of securitization likelihood

$$Pr(\text{Securitized}_{i,t} = 1) = \Phi(\alpha + \beta \text{FacilityType}_{i,t} + \gamma \text{BorrowerFactors}_{i,t} + \phi \text{MacroFactors}_t). \quad (2)$$

Interpretation: This specification validates the empirical proxies for securitization-friendliness. If Term Loan B, B-rated, and securitization-active-bank variables do not predict actual securitization, the later pricing tests would be less convincing.

4.3 Term Loan B spread test

$$\text{Spread}_{i,t} = \alpha + \beta_1(\text{SecActive}_{j,t} \times \text{BRated}_{i,t}) + \beta_2 \text{SecActive}_{j,t} + \beta_3 \text{BRated}_{i,t} + \delta' X_{i,t} + FE + \varepsilon_{i,t}. \quad (3)$$

Interpretation: The coefficient β_1 captures whether B-rated Term Loan B facilities originated by securitization-active banks receive an incremental spread discount. This is the paper's main collateral-demand pricing test.

4.4 Pooled facility-type triple-interaction test

$$\text{Spread}_{i,t} = \alpha + \theta(\text{TermB}_i \times \text{SecActive}_j \times \text{BRated}_i) + \text{lower-order terms} + \delta' X_{i,t} + FE + \varepsilon_{i,t}. \quad (4)$$

Interpretation: The coefficient θ tests whether the securitization-friendly discount is specific to Term Loan Bs rather than appearing in Term Loan As or revolvers with similar ratings and arrangers. This helps separate CLO collateral demand from a general bank risk-tolerance story.

4.5 Within-loan spread-difference tests

$$\text{Spread}_d^{\text{TermB}} - \text{Spread}_d^{\text{TermA}} = \alpha + \beta(\text{SecActive}_d \times \text{BRated}_d) + \delta' X_d + FE + u_d, \quad (5)$$

$$\text{Spread}_d^{\text{TermB}} - \text{Spread}_d^{\text{Revolver}} = \alpha + \beta(\text{SecActive}_d \times \text{BRated}_d) + \delta' X_d + FE + u_d. \quad (6)$$

Interpretation: These are the cleanest tests in the paper. Facilities in the same deal share the same borrower and many deal-level risks, so differences in spreads are more plausibly tied to facility-specific collateral value.

4.6 CLO ownership by rating

$$\text{CLOOwnership}_r = \frac{\text{Amount of facility held by CLOs in rating group } r}{\text{Total facility amount in rating group } r}. \quad (7)$$

Interpretation: The ownership calculations test whether CLOs actually hold more of the debt categories predicted by the pricing results, especially lower-rated Term Loan Bs.

5 Identification and empirical design

5.1 Core identification problem

The key challenge is that securitized loans might have lower spreads because CLO demand lowers the cost of capital, but they might also have lower spreads because borrowers, banks, or loan terms differ in unobserved ways. The paper addresses this problem through a sequence of increasingly targeted tests.

5.2 Step 1: direct comparison

The first step compares securitized and non-securitized loans conditional on controls and fixed effects. This establishes the basic pricing pattern: securitized facilities have lower spreads. However, because actual securitization is endogenous and imperfectly measured, this evidence is not decisive.

5.3 Step 2: validating securitization-friendly characteristics

The second step shows that Term Loan B facilities, B-rated facilities, and facilities arranged by securitization-active banks are more likely to be securitized. This is important because the later regressions use these characteristics to proxy for CLO demand.

5.4 Step 3: triple interaction

The third step tests whether the discount is concentrated in the exact combination most valued by CLOs: Term Loan B, B-rated borrower, securitization-active bank. If results were caused only by risk tolerance of securitization-active banks, similar discounts should appear for Term Loan As and revolvers. The data show the opposite.

5.5 Step 4: within-loan comparisons

The fourth step compares spreads across facilities inside the same loan. This is powerful because Term Loan B and Term Loan A/revolver facilities often finance the same borrower at the same time. The within-loan tests hold borrower risk and deal timing almost fixed.

5.6 Remaining limitations

- Moody's CLO collateral data are a snapshot as of 2009, so some securitized loans are missed.
- Securitization-active banks may differ from other banks in unobserved underwriting or relationship strategies.
- The paper cannot directly observe the bargaining process between borrowers, banks, and CLO investors.
- Loan covenants and non-price terms may also adjust, so spread discounts are only part of the full contract effect.

Interpretation: The identification is not a randomized experiment, but the layered evidence strongly narrows the set of alternative explanations.

6 Section-by-section study guide

- **Introduction.** Sets up the puzzle: CLOs and leveraged loans grew dramatically from 2002 to 2007, but whether securitization affected primary-market corporate borrowing costs was unclear.
- **Institutional background.** Explains why Term Loan Bs, B-rated loans, and securitization-active banks matter for CLO collateral demand.
- **Data.** Describes the Dealscan loan sample, the Moody's CLO collateral data, ratings screens, and construction of the securitization-active bank variable.
- **Results: direct pricing.** Shows that securitized loans have lower spreads than non-securitized loans after controls.

- **Results: securitization likelihood.** Confirms that the proposed securitization-friendly characteristics predict actual CLO inclusion.
- **Results: collateral-demand tests.** Shows that the spread discount is concentrated in B-rated Term Loan B facilities originated by securitization-active banks.
- **Within-loan tests.** Compares facilities in the same deal to reduce borrower-level confounding.
- **Alternative explanations.** Discusses LBO demand and broader institutional demand, then uses rating-level CLO ownership data to argue CLO demand is an important driver.
- **Conclusion.** Interprets the evidence as consistent with securitization lowering corporate debt costs during the CLO boom.

7 Tables and figures

7.1 Figure 1: Plots of Term Loan B origination volume

Read:

- Panel A compares Term Loan B originations by securitization-active and non-securitization-active banks.
- The two series look broadly similar before 2002, then securitization-active banks expand Term Loan B originations much more aggressively.
- Panel B shows that within securitization-active banks, the boom is especially strong for B-rated Term Loan Bs.
- Panel C shows that non-securitization-active banks also increase B-rated Term Loan B origination, but the rise is smaller.

Detailed interpretation: The figure is the visual foundation for the empirical strategy. It shows that the securitization boom is not merely an overall leveraged-loan boom. The rise is strongest exactly where the CLO-demand mechanism predicts: among banks active in CLO underwriting and in lower-rated Term Loan B facilities.

Economic message: CLO demand appears to have shifted the behavior of banks that could use loans as collateral in structured products.

7.2 Table 1: Facility summary statistics by facility type, credit rating, facility purpose, and facility attribute

Read:

- Panel A compares securitized and non-securitized facilities across revolvers, Term Loan A, and Term Loan B facilities.
- Securitized Term Loan Bs have lower mean spreads than non-securitized Term Loan Bs.
- Panel B shows that within Term Loan Bs, securitized facilities have lower spreads in every rating category from Ba1 to B3.
- The spread difference is especially large in lower credit categories, reaching 73.1 basis points in B3 facilities.
- Panel D shows securitized facilities are larger, longer maturity, more likely to have covenants, and arranged in smaller syndicates.

Detailed interpretation: Table 1 provides the raw facts that motivate the rest of the paper. The pricing differences are large, but the facility differences also show why simple comparisons are not enough. Securitized loans differ in amount, maturity, covenants, and syndicate structure, so regressions with controls are necessary.

Economic message: Securitized loans look cheaper, but they are not randomly selected. The paper's later tests ask whether the cheaper spreads reflect collateral demand rather than selection.

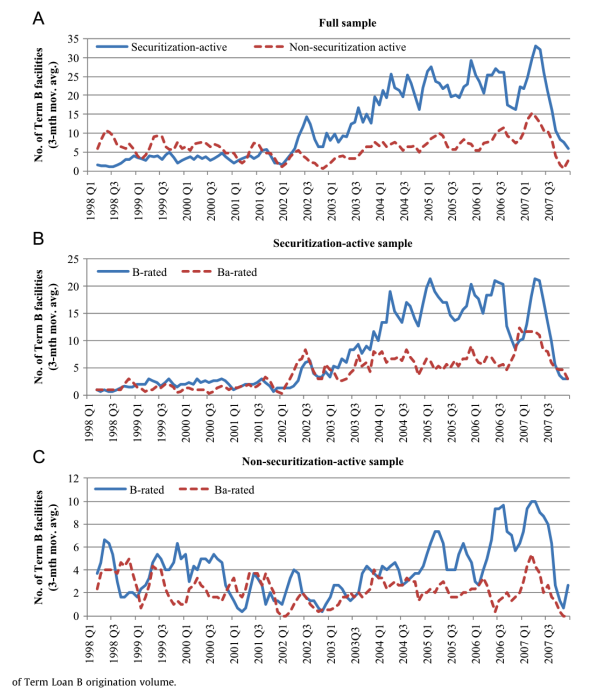


Figure 1: Plots of Term Loan B origination volume.

Table 1
Facility summary statistics by facility type, credit rating, facility purpose, and facility attribute.

In this table we report summary statistics on loan facilities identified as having been securitized and compare them against the attributes of non-securitized facilities. The Moody's EMS database provides a "snapshot" of the characteristics and identity of all the securities serving as collateral in any Moody's-rated CLO as of the first quarter of 2008. The securitized sample is defined as any loan facility identified as collateral in a CLO according to Moody's EMS database as of the first quarter of 2008. We outline the matching process in the text. We report summary statistics on the number of loan facilities and all-in-drawn spread over LIBOR charged on loan facilities at the time of loan origination. Panels A-C report summary statistics by facility type, credit rating, and facility purpose, respectively. Panel D reports summary statistics on facility attributes. The DealScan sample consists of senior secured and unsecured loan facilities originated between 2002 and September 2007 which have Moody's credit ratings on the senior debt of the issuing firm available at the time of loan origination. We restrict our sample to include syndicated loan facilities originated in the United States and exclude firms with SIC codes between 6000 and 6500. We also restrict our analysis to facilities with senior debt rated Ba1, Ba2, Ba3, B1, B2, and B3. We identify and remove any second-lien or mezzanine facilities from the sample.

Panel A Non-securitized sample							Securitized sample							
Facility type	Count	All-in-drawn spread over LIBOR				75th tile	Count	All-in-drawn spread over LIBOR				75th tile	Diff	t-Stat
		25th tile	Median	Mean	75th tile			25th tile	Median	Mean	75th tile			
Revolver	2514	150.0	225.0	226.8	275.0	34	200.0	250.0	235.3	250.0	8.5	(1.06)		
Term Loan A	278	150.0	225.0	228.0	300.0	20	143.8	150.0	175.6	200.0	-52.4	(-3.40)		
Term Loan B	1413	200.0	250.0	276.7	325.0	277	200.0	225.0	240.4	275.0	-36.3	(-7.39)		
Panel B							Term Loan Bs only - securitized sample							
Term Loan Bs only - Non-securitized sample							All-in-drawn spread over LIBOR							
Moody's sr. debt credit rating	Count	25th tile	Median	Mean	75th tile	Count	25th tile	Median	Mean	75th tile	Diff	t-Stat		
Ba1	49	175.0	200.0	224.5	275.0	9	175.0	175.0	181.9	187.5	-42.5	(-2.86)		
Ba2	138	175.0	225.0	230.7	275.0	22	150.0	187.5	200.0	225.0	-30.7	(-2.13)		
Ba3	290	200.0	225.0	250.0	275.0	50	175.0	225.0	223.0	250.0	-27.0	(-2.24)		
B1	458	225.0	250.0	273.9	300.0	101	200.0	225.0	241.1	275.0	-32.8	(-4.67)		
B2	338	225.0	275.0	306.6	350.0	68	225.0	250.0	265.4	287.5	-41.2	(-3.39)		
B3	140	250.0	275.0	332.3	400.0	27	225.0	250.0	259.3	275.0	-73.1	(-4.27)		
Panel C							Term Loan Bs only--securitized sample							
Term Loan Bs only--Non-securitized sample							All-in-drawn spread over LIBOR							
Facility purpose	Count	25th tile	Median	Mean	75th tile	Count	25th tile	Median	Mean	75th tile	Diff	t-Stat		
Corporate purposes	445	200.0	250.0	267.9	325.0	85	200.0	225.0	241.2	250.0	-26.8	(-2.65)		
Debt repayment	123	225.0	275.0	308.5	350.0	30	175.0	200.0	214.2	250.0	-94.3	(-6.56)		
Takeover	183	200.0	225.0	250.9	275.0	39	200.0	200.0	227.2	250.0	-23.6	(-2.14)		
Working capital	168	200.0	250.0	279.7	325.0	19	175.0	225.0	226.3	275.0	-53.4	(-3.18)		
Leveraged buyout (LBO)	209	225.0	275.0	289.1	325.0	63	225.0	250.0	256.7	300.0	-32.4	(-4.04)		
Acquisition line	112	218.8	250.0	266.3	300.0	12	175.0	225.0	256.3	337.5	-10.0	(-0.39)		
Other	173	225.0	267.5	277.4	304.2	29	200.0	219.6	243.1	264.3	-	-		
Panel D							Term Loan Bs only--securitized sample							
Term Loan Bs only--Non-securitized sample							All-in-drawn spread over LIBOR							
Loan attribute	Count	25th tile	Median	Mean	75th tile	Count	25th tile	Median	Mean	75th tile	Diff	t-Stat		
Facility amount (\$ million)	1432	\$125.0	\$219.5	\$388.0	\$400.0	278	\$180.0	\$315.0	\$519.8	\$600.0	\$131.8	(2.91)		
Sales size (\$ million)	1432	\$272.7	\$616.5	\$1,757.3	\$1,500.7	278	\$347.3	\$704.1	\$1,055.0	\$1,510.7	\$1,297.6	(1.58)		
Tranche maturity (months)	1355	60.0	72.0	71.4	84.0	276	72.0	84.0	76.3	84.0	5.0	(5.74)		
Syndicate size (#of participants)	1431	3.0	5.0	8.5	9.0	277	3.0	5.0	7.0	7.0	-1.4	(-2.37)		
% With covenants	1432	-	-	54.2%	-	278	-	-	62.6%	-	8.40%	(2.63)		

Table 1: Facility summary statistics by facility type, credit rating, facility purpose, and facility attribute.

7.3 Table 2: The creation of CLOs through time and by originator

Read:

- CLO origination volume rises sharply after 2002 and peaks in 2006–2007.
- Total CLO counts rise from 43 in 2002 to more than 200 in 2006 and 2007.
- The top ten CLO underwriting banks account for a large share of deals, motivating the securitization-active classification.
- The largest originators include JPMorgan, Citigroup, Bear Stearns, Credit Suisse, Lehman Brothers, Deutsche Bank, Goldman Sachs, Morgan Stanley, Wachovia, and Merrill Lynch.

Detailed interpretation: Table 2 establishes that CLO underwriting is concentrated. This concentration makes bank identity informative: if a lead arranger is among the major CLO underwriters, it is plausibly better positioned to originate loans for CLO collateral.

Economic message: The supply of securitizable loans is linked to intermediary structure. Banks with CLO underwriting platforms had stronger incentives to originate collateral-friendly loans.

7.4 Table 3: Do securitized loans have lower spreads?

Read:

- The dependent variable is the all-in-drawn spread.
- The main coefficient is the securitized indicator.
- In the fully controlled B-rated Term Loan B specification, securitized loans have spreads about 17.7 basis points lower.
- Results are robust to fixed effects for purpose, year, lead arranger, industry, and credit rating where appropriate.

Detailed interpretation: Table 3 provides direct evidence that loans actually appearing in CLOs are cheaper at origination. However, actual securitization is not randomly assigned and is imperfectly observed, so the table is best viewed as the first piece of evidence rather than the final causal estimate.

Economic message: Facilities that are ultimately securitized are priced at a discount, consistent with CLO demand lowering borrowing costs.

CLO originating bank	1996	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006	2007	2008	Total
JPMorgan	0	1	0	3	2	6	6	7	7	13	26	18	5	94
Citigroup	1	0	0	1	0	0	0	2	1	17	28	24	11	85
Bear, Stearns & Co. Inc.	0	0	0	4	2	3	2	3	12	13	21	17	1	78
Credit Suisse	0	0	1	2	2	2	7	10	5	12	13	14	1	69
Lehman Brothers	1	0	2	2	1	1	3	3	6	11	10	19	5	64
Deutsche Bank	0	0	1	1	0	4	3	8	5	2	16	18	5	63
Goldman Sachs & Co.	0	0	1	2	5	5	4	5	3	6	12	17	3	63
Morgan Stanley	0	1	2	3	3	3	3	2	3	7	16	18	2	63
Wachovia Securities Inc.	0	0	1	2	2	0	3	3	5	9	18	18	2	63
Merrill Lynch & Co.	1	2	4	1	3	1	2	2	4	10	7	10	3	50
Bank of America Securities LLC	0	0	0	1	1	1	1	2	2	5	14	7	3	37
UBS Securities LLC	0	0	0	0	0	1	0	1	3	5	5	5	1	21
IXIS Securities North America Inc.	0	0	0	0	0	0	0	2	3	2	9	1	0	17
Barclays Capital	0	0	0	0	0	0	1	1	0	0	2	4	12	12
CIBC World Markets Inc.	0	2	2	3	1	0	2	0	0	0	0	0	0	10
Royal Bank of Scotland plc	0	0	0	0	0	0	0	0	0	0	1	7	0	8
Salomon Smith Barney	0	0	0	0	2	1	3	2	0	0	0	0	0	8
BNP Paribas	0	0	0	0	0	0	0	1	0	1	2	3	0	7
Others (28 other banks)	0	0	2	4	7	1	3	2	5	10	20	13	5	72
Total	3	6	16	29	31	29	43	56	64	123	220	213	51	884

Table 2: The creation of CLOs through time and by originator.

Table 3
Do securitized loans have lower spreads?
In this table we estimate an OLS regression of facility attributes and other controls on the all-in-drawn spread of loan facilities originated between 2002 and September 2007. We create an indicator variable, *Securitized indicator*, for loans identified as collateral in a CLO as of 2009 using Moody's EMS database. *Term structure* is calculated as the difference between the five-year and three-month T-bill rate in the month of deal origination. *Credit spread* is the spread of a 10-year Ba- and B-rated bond index over the 10-year T-bond rate, respectively. *Covenants indicator* is equal to one if a facility has financial covenants. *Pricing grid indicator* is equal to one if the spread is adjustable based on accounting performance. *Size of syndicate* is the number of lenders involved in the lending syndicate. *Tranche maturity* is the number of months until the facility matures. *Log facility amount* and *Log sales size* represent the dollar amount of the loan facility and total sales of the borrowing firm, measured in U.S. \$ million. *Secured indicator* is an indicator equal to one if the loan facility is secured. In various specifications, as indicated in the table, we control for tranche-type fixed-effects (Revolver, Term Loan A or Term Loan B), credit-rating fixed-effects (Ba1, Ba2, Ba3, B1, B2, B3), tranche-purpose fixed-effects ("corporate purposes," "LBO," "debt repayment," etc.), year fixed-effects, lead arranger fixed-effects, and industry fixed-effects (two-digit SIC code). We restrict our sample to include syndicated loan facilities originated in the United States and exclude firms with SIC codes between 6000 and 6500. We also restrict our analysis to loan facilities with senior debt rated Ba1, Ba2, Ba3, B1, B2, and B3. We identify and remove any second-lien or mezzanine facilities from the sample. Standard errors are clustered by year and by borrowing firm. The symbols ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Dependent var.: All-in-drawn spread	Term B only, Ba and B-rated sr. debt	Term B only, Ba-rated sr. debt	Term B only, B-rated sr. debt
	(1)	(2)	(3)
Securitized indicator	-6.872 (1.46)	16.476*** (3.00)	-17.698** (2.07)
Term structure	3.214 (0.24)	0.931 (0.06)	7.714 (0.80)
Credit spread	32.857*** (3.15)	44.273*** (3.84)	21.228* (1.83)
Covenants indicator	-9.372 (1.35)	-19.570*** (2.98)	-8.264 (1.04)
Pricing grid indicator	-8.665 (1.15)	-3.492 (0.43)	-12.308 (1.52)
Size of syndicate	-0.125** (0.05)	-0.212 (0.06)	-0.066*** (0.02)
Tranche maturity	-1.133*** (0.26)	-1.083*** (0.36)	-1.257* (0.59)
Log facility amount	-0.743 (0.29)	-0.743 (0.29)	-1.1715*** (0.15)
Log sales size	-3.828* (1.75)	-5.024* (1.74)	-1.422 (0.47)
Secured indicator	2.743 (0.48)	-10.052 (0.92)	13.352 (1.04)
Constant	334.458*** (6.49)	302.778*** (4.00)	621.844*** (9.28)
Tranche-type fixed effects	No	No	No
Moody's sr. debt fixed effects	Yes	Yes	No
Tranche-purpose fixed effects	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes
Lead arranger fixed effects	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes
Standard errors clustered by:			
Borrower	Yes	Yes	Yes
Year	Yes	Yes	Yes
Adjusted R ²	0.386	0.482	0.312
Observations	1613	540	1073

Table 3: Do securitized loans have lower spreads?

7.5 Table 4: What type of loans get securitized?

Read:

- Term Loan B facilities are more likely to be securitized than Term Loan A facilities.
- Revolvers are much less likely to be securitized.
- Facilities originated by securitization-active banks are more likely to enter CLOs.
- B-rated senior debt is more likely to be securitized than Ba-rated debt in the relevant specifications.
- Larger facilities and secured facilities are also more likely to appear in CLOs.

Detailed interpretation: This table validates the proxies used in the pricing tests. The paper does not simply assume which loans CLOs like; it verifies that these characteristics actually predict inclusion in CLO collateral pools.

Economic message: CLOs do not demand all loans equally. Their demand is concentrated in facility structures and credit qualities that are easy to tranche and repackage.

Table 4
What type of loans get securitized?
In columns 1 and 2 we estimate a probit model where the dependent variable is equal to one if the loan facility is identified as collateral in a CLO as of 2009, and zero otherwise. Column 3 employs the same dependent variable but the coefficients are estimated using an OLS linear probability model. Term Loan B is equal to one if the facility is a Term Loan B facility. Securitization active bank is equal to one if any of the lead arrangers of the loan facility are among the top-ten CLO underwriters (see Table 2). B-rated senior debt is equal to one if the senior debt associated with the loan facility is rated B1, B2, or B3 at loan origination. Revolver is equal to one if the loan facility is a revolving facility (Term Loan As serve as the omitted loan facility type). Covenants indicator is equal to one if a facility has financial covenants. Pricing grid indicator is equal to one if the spread is adjustable based on accounting performance. Syndicate size is the number of lenders involved in the lending syndicate. Log facility amount and Log sales size represent the dollar amount of the loan facility and total sales of the borrowing firm, measured in U.S. million. Secured indicator is an indicator equal to one if the loan facility is secured. In various specifications, as indicated in the table, we control for tranche-purpose fixed-effects ("corporate purposes," "LBO," "debt repayment," etc.), year fixed-effects, and industry fixed-effects (two-digit SIC code). We restrict our sample to include syndicated loan facilities originated in the United States and exclude firms with SIC codes between 6000 and 6500. We also restrict our analysis to loan facilities with senior debt rated Baa1, Baa2, Baa3, B1, B2, and B3. We identify and remove any second-lien or mezzanine facilities from the sample. The sample size in column 2 drops slightly because the inclusion of fixed effects perfectly predicts outcomes in the left-hand side variable. These observations are dropped from the estimation. Standard errors are clustered by year. The symbols ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Dependent var. = 1 if loan is identified as having been securitized	Estimated using probit		Estimated using OLS (Mean dep. var. = 0.071)
	Term B, Term A, and Revolvers (1)	Term B, Term A, and Revolvers (2)	Term B, Term A, and Revolvers (3)
Term Loan B*B-rated senior debt			0.018* (1.62)
Term Loan B*Securitization-active bank			0.038* (1.94)
Term Loan B	0.341*** (3.128)	0.484*** (3.966)	0.030*** (2.92)
B-rated senior debt	0.156** (1.987)	0.126 (1.440)	0.004 (0.80)
Securitization-active bank	0.218*** (2.691)	0.301*** (3.387)	0.006 (0.93)
Revolver	-0.874*** (-7.327)	-0.907*** (-7.207)	-0.065*** (8.55)
Covenants indicator	0.069 (0.796)	0.300*** (3.157)	0.036*** (3.29)
Pricing grid indicator	0.076 (0.910)	0.070 (0.771)	-0.003 (0.17)
Syndicate size	-0.01*** (-2.636)	-0.005 (-1.418)	-0.001* (1.95)
Log facility amount	0.158*** (4.096)	0.067 (1.553)	0.015*** (6.79)
Log sales size	0.008 (0.256)	0.010 (0.270)	-0.002 (0.24)
Secured indicator	0.522*** (5.139)	0.452*** (4.333)	0.037*** (5.40)
Constant	-5.222*** (-7.229)	-4.835*** (-4.909)	-0.402*** (3.92)
Tranche-purpose fixed effects	No	Yes	Yes
Year fixed effects	No	Yes	Yes
Industry fixed effects	No	Yes	Yes
Standard errors clustered by:			
Borrower	Yes	Yes	Yes
Credit rating	No	No	Yes
Pseudo-R ²	0.201	0.297	
Adjusted R ²			0.124
Observations	5101	4755	5070

Table 4: What type of loans get securitized?

Table 5
Do "securitization-friendly" Term Loan B facilities have lower spreads?
In this table we estimate an OLS regression of facility attributes and other controls on the all-in-drawn spread of loan facilities originated between 2002 and September 2007. Securitization active bank is equal to one if any of the lead arrangers of the loan facility are among the top-ten CLO underwriters (see Table 2). B-rated senior debt is equal to one if the senior debt associated with the loan facility is rated B1, B2, or B3 at loan origination. Term structure is calculated as the difference between the 5-year and 3-month T-bill rate in the month of deal origination. Credit spread is the spread of a 10-year Baa- and B-rated bond index over the 10-year T-bond rate, respectively. Covenants indicator is an indicator equal to one if a facility has financial covenants. Pricing grid indicator is equal to one if the spread is adjustable based on accounting performance. Size of syndicate is the number of lenders involved in the lending syndicate. Tranche maturity is the number of months until the facility matures. Log facility amount and Log sales size represent the dollar amount of the loan facility and total sales of the borrowing firm, measured in U.S. \$ million. Secured indicator is an indicator equal to one if the loan facility is secured. In various specifications, as indicated in the table, we control for credit-rating fixed-effects (Baa1, Baa2, Baa3, B1, B2, B3), tranche-purpose fixed-effects ("corporate purposes," "LBO," "debt repayment," etc.), year fixed-effects, and industry fixed-effects (two-digit SIC code). We restrict our sample to include syndicated loan facilities originated in the United States and exclude firms with SIC codes between 6000 and 6500. We also restrict our analysis to loan facilities with senior debt rated Baa1, Baa2, Baa3, B1, B2, and B3. We identify and remove any second-lien or mezzanine facilities from the sample. Standard errors are clustered by year and by borrowing firm. The symbols ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Dependent var.: All-in-drawn spread	Term Loan B sample (1)	Term Loan B sample (2)	Term Loan B sample (3)
Securitization-active bank*B-rated sr. debt			-20.918*** (2.65)
Securitization-active bank	-11.473* (1.65)	-11.337 (1.55)	2.801 (0.34)
B-rated sr. debt		20.924** (2.39)	36.366*** (5.37)
Term structure	4.870 (0.35)	5.412 (0.43)	5.406 (0.44)
Credit spread	33.965*** (2.90)	32.985*** (3.00)	33.239*** (3.02)
Covenants indicator	-11.133 (1.56)	-15.189** (1.97)	-15.153** (2.00)
Pricing grid indicator	-9.730 (1.32)	-10.454 (1.24)	-10.246 (1.22)
Size of syndicate	-0.351** (2.34)	-0.415*** (2.69)	-0.414*** (2.80)
Tranche maturity	-1.194*** (2.92)	-1.284*** (3.05)	-1.280*** (3.04)
Log facility amount	-8.142*** (2.21)	-9.119*** (2.91)	-9.153*** (2.93)
Log sales size	-5.272** (1.863)	-4.045 (1.43)	-4.082 (1.46)
Secured indicator	4.697 (0.30)	4.697 (0.59)	4.794 (0.60)
Constant	421.483*** (7.20)	440.514*** (5.36)	655.358*** (6.68)
Moody's sr. debt fixed effects	Yes	No	No
Tranche-purpose fixed effects	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes
Lead arranger fixed effects	No	No	No
Standard errors clustered by			
Borrower	Yes	Yes	Yes
Year	Yes	Yes	Yes
Adjusted R ²	0.372	0.343	0.344
Observations	1613	1613	1613

Table 5: Do securitization-friendly Term Loan B facilities have lower spreads?

7.6 Table 5: Do securitization-friendly Term Loan B facilities have lower spreads?

Read:

- The sample is restricted to Term Loan B facilities from 2002 through September 2007.
- Column 3 includes the key interaction between securitization-active bank and B-rated senior debt.
- The interaction coefficient is about -20.9 basis points and statistically significant.
- The standalone securitization-active coefficient is positive when the interaction is included, meaning the discount is concentrated in B-rated loans rather than all loans from securitization-active banks.

Detailed interpretation: Table 5 is one of the central tests. It shows that the lower spread is not a general bank effect. The discount appears when the arranger is securitization-active and the borrower is B-rated, matching the collateral-demand mechanism.

Economic message: Banks appear willing to price B-rated Term Loan Bs more cheaply when those loans are especially useful as CLO collateral.

7.7 Table 6: Is the securitization-friendly decline in spreads specific to Term Bs only?

Read:

- The table pools Term Loan Bs with either Term Loan As or revolvers.
- The key triple interaction is Term Loan B $\times$ securitization-active bank $\times$ B-rated senior debt.
- Relative to Term Loan As, the incremental Term Loan B discount is about 30.6 basis points.
- Relative to revolvers, the incremental discount is about 20.8 basis points.
- Similar discounts do not appear for Term Loan As or revolvers with securitization-friendly ratings and arrangers.

Detailed interpretation: Table 6 directly addresses the alternative story that securitization-active banks simply became more aggressive lenders. If that were true, discounts should occur for all facilities from these banks. Instead, the discount is tied to Term Loan B structure, which is what CLOs prefer.

Economic message: The pricing effect is facility-specific, not just bank-specific. That strongly supports a collateral-demand interpretation.

Table 6
Is the "securitization-friendly" decline in spreads specific to Term Bs only?

In this table we estimate an OLS regression of facility attributes and other controls on the all-in-drawn spread of loan facilities originated between 2002 and September 2007. *Term Loan B* is equal to one if the facility is a Term Loan B facility. *Securitization active bank* is equal to one if any of the lead arrangers of the loan facility are among the top-ten CLO underwriters (see Table 2). *B-rated senior debt* is equal to one if the senior debt associated with the loan facility is rated B1, B2, or B3 at loan origination. *Term structure* is calculated as the difference between the 5-year and 3-month T-bill rate in the month of deal origination. *Credit spread* is the spread of a 10-year Ba- and B-rated bond index over the 10-year T-bond rate, respectively. *Covenants indicator* is equal to one if a facility has financial covenants. *Pricing grid indicator* is equal to one if the spread is adjustable based on accounting performance. *Size of syndicate* is the number of lenders involved in the lending syndicate. *Tranche maturity* is the number of months until the facility matures. *Log facility amount* and *Log sales size* represent the dollar amount of the loan facility and total sales of the borrowing firm, measured in U.S. \$ million. *Secured indicator* is an indicator equal to one if the loan facility is secured. We control for tranche-purpose fixed-effects ("corporate purposes," "LBO," "debt repayment," etc.), year fixed-effects, and industry fixed-effects (two-digit SIC code). We restrict our sample to include syndicated loan facilities originated in the United States and exclude firms with SIC codes between 6000 and 6500. We also restrict our analysis to loan facilities with senior debt rated Ba1, Ba2, Ba3, B1, B2, and B3. We identify and remove any second-lien or mezzanine facilities from the sample. Standard errors are clustered by year and by borrowing firm. The symbols ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Dependent var.: All-in-drawn spread	Term Loan B and Term Loan A sample (1)	Term Loan B and Revolver sample (2)
Term Loan B*Securitization-active bank*B-rated sr. debt	-30.606** (2.03)	-20.761** (2.27)
Term Loan B*Securitization-active bank	17.350 (1.14)	-7.571 (1.31)
Term Loan B*B-rated sr. debt	12.008 (0.95)	9.085 (0.92)
Securitization-active bank*B-rated sr. debt	0.027 (0.44)	-7.284* (1.70)
Term Loan B	40.230** (2.76)	52.713** (11.41)
Securitization-active bank	-13.681 (0.77)	14.893*** (2.60)
B-rated sr. debt	23.407* (1.70)	33.746*** (4.18)
Term structure	4.965 (0.40)	7.719 (1.13)
Credit spread	34.684*** (2.76)	28.186*** (4.52)
Covenants indicator	-14.973** (2.11)	-12.736* (1.89)
Pricing grid indicator	-10.560* (1.65)	-17.318*** (3.21)
Size of syndicate	-0.410*** (3.07)	-0.536*** (3.91)
Tranche maturity	-1.252*** (3.16)	-1.252*** (3.16)
Log facility amount	-8.700*** (3.22)	-20.518*** (7.67)
Log sales size	-4.587 (1.64)	0.246 (0.10)
Secured indicator	4.440 (0.70)	16.044*** (3.41)
Constant	548.103*** (5.12)	538.217*** (12.37)
Tranche-purpose fixed effects	Yes	Yes
Year fixed effects	Yes	Yes
Industry fixed effects	Yes	Yes
Standard errors clustered by:		
Borrower	Yes	Yes
Year	Yes	Yes
Adjusted R ²	0.408	0.428
Observations	1904	4238

Table 6
Is the "securitization-friendly" decline in spreads specific to Term Bs only?

In this table we estimate an OLS regression of facility attributes and other controls on the all-in-drawn spread of loan facilities originated between 2002 and September 2007. *Term Loan B* is equal to one if the facility is a Term Loan B facility. *Securitization active bank* is equal to one if any of the lead arrangers of the loan facility are among the top-ten CLO underwriters (see Table 2). *B-rated senior debt* is equal to one if the senior debt associated with the loan facility is rated B1, B2, or B3 at loan origination. *Term structure* is calculated as the difference between the 5-year and 3-month T-bill rate in the month of deal origination. *Credit spread* is the spread of a 10-year Ba- and B-rated bond index over the 10-year T-bond rate, respectively. *Covenants indicator* is equal to one if a facility has financial covenants. *Pricing grid indicator* is equal to one if the spread is adjustable based on accounting performance. *Size of syndicate* is the number of lenders involved in the lending syndicate. *Tranche maturity* is the number of months until the facility matures. *Log facility amount* and *Log sales size* represent the dollar amount of the loan facility and total sales of the borrowing firm, measured in U.S. \$ million. *Secured indicator* is an indicator equal to one if the loan facility is secured. We control for tranche-purpose fixed-effects ("corporate purposes," "LBO," "debt repayment," etc.), year fixed-effects, and industry fixed-effects (two-digit SIC code). We restrict our sample to include syndicated loan facilities originated in the United States and exclude firms with SIC codes between 6000 and 6500. We also restrict our analysis to loan facilities with senior debt rated Ba1, Ba2, Ba3, B1, B2, and B3. We identify and remove any second-lien or mezzanine facilities from the sample. Standard errors are clustered by year and by borrowing firm. The symbols ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Dependent var.: All-in-drawn spread	Term Loan B and Term Loan A sample (1)	Term Loan B and Revolver sample (2)
Term Loan B*Securitization-active bank*B-rated sr. debt	-30.606** (2.03)	-20.761** (2.27)
Term Loan B*Securitization-active bank	17.350 (1.14)	-7.571 (1.31)
Term Loan B*B-rated sr. debt	12.008 (0.95)	9.085 (0.92)
Securitization-active bank*B-rated sr. debt	0.027 (0.44)	-7.284* (1.70)
Term Loan B	40.230** (2.76)	52.713** (11.41)
Securitization-active bank	-13.681 (0.77)	14.893*** (2.60)
B-rated sr. debt	23.407* (1.70)	33.746*** (4.18)
Term structure	4.965 (0.40)	7.719 (1.13)
Credit spread	34.684*** (2.76)	28.186*** (4.52)
Covenants indicator	-14.973** (2.11)	-12.736* (1.89)
Pricing grid indicator	-10.560* (1.65)	-17.318*** (3.21)
Size of syndicate	-0.410*** (3.07)	-0.536*** (3.91)
Tranche maturity	-1.252*** (3.16)	-1.252*** (3.16)
Log facility amount	-8.700*** (3.22)	-20.518*** (7.67)
Log sales size	-4.587 (1.64)	0.246 (0.10)
Secured indicator	4.440 (0.70)	16.044*** (3.41)
Constant	548.103*** (5.12)	538.217*** (12.37)
Tranche-purpose fixed effects	Yes	Yes
Year fixed effects	Yes	Yes
Industry fixed effects	Yes	Yes
Standard errors clustered by:		
Borrower	Yes	Yes
Year	Yes	Yes
Adjusted R ²	0.408	0.428
Observations	1904	4238

Table 6: Is the securitization-friendly decline in spreads specific to Term Bs only?

Table 7: Within-deal differences in loan spreads: Term Bs versus Term As.

7.8 Table 7: Within-deal differences in loan spreads: Term Bs versus Term As

Read:

- The dependent variable is the spread difference between Term Loan B and Term Loan A facilities in the same deal.

- The key securitization-active bank $\times$ B-rated interaction is about -47.8 basis points in column 3.
- The result implies that when the same borrower issues both Term Loan B and Term Loan A facilities, the Term Loan B is relatively cheaper when it is more attractive as CLO collateral.
- Observations fall to 186 deals because the test requires loans with both facility types.

Detailed interpretation: This is a high-value identification test. Comparing facilities inside the same deal controls for borrower credit quality, origination date, and many deal-level factors. The remaining difference is more plausibly tied to facility-specific collateral value.

Economic message: CLO demand lowers the relative spread of the facility type that CLOs want, even when compared to another facility issued by the same borrower at the same time.

7.9 Table 8: Within-deal differences in loan spreads: Term Bs versus revolvers

Read:

- The dependent variable is the spread difference between Term Loan B and revolver facilities in the same deal.
- The securitization-active bank $\times$ B-rated interaction is negative in column 3, about -14.2 basis points, and statistically significant at the 10% level.
- The sample is larger than the Term Loan A comparison because more deals include revolvers.
- The result is weaker than Table 7 but points in the same direction.

Detailed interpretation: Revolvers have optional drawdowns and different expected maturities, so they are less clean collateral for CLOs. The fact that Term Loan B discounts appear relative to revolvers reinforces the interpretation that CLO collateral demand is tied to structure, not just borrower risk.

Economic message: Facility cash-flow structure matters for securitization value. CLO demand shows up most clearly in bullet-like Term Loan Bs.

7.10 Table 9: Differences in CLO ownership by credit rating

Read:

- Panel A reports total CLO ownership percentages of Term Loan Bs by rating class.
- Panel B reports average CLO ownership percentages by rating class.
- CLOs hold larger total and average shares of lower-rated Term Loan Bs than higher-rated Term Loan Bs.
- The strongest total-ownership difference appears when B2 and B3 loans are compared with Ba1 and Ba2 loans: the difference is 7.67 percentage points.
- The average ownership differences are smaller but directionally consistent.

Detailed interpretation: Table 9 addresses the concern that the spread results might reflect general institutional demand rather than CLO demand specifically. The table shows that CLO ownership is heavier in lower-rated loans, which are exactly where the pricing discount is strongest.

Economic message: The pricing evidence and the ownership evidence line up: CLOs disproportionately own lower-rated Term Loan B facilities, and those facilities receive spread discounts when originated by securitization-active banks.

Table 8

Within-deal differences in loan spreads: Term Bs vs. revolvers.

In this table we estimate an OLS regression of facility attributes and other controls on differences in the all-in-drawn spread of Term Loan B facilities compared to revolving facilities within the same deal. The sample size of 1,150 represents the number of deals that have both Revolvers and Term Loan B facilities during our 2002–September 2007 sample period as well as data on the required control variables. *Securitization active bank* is equal to one if any of the lead arrangers of the loan facility are among the top-ten CLO underwriters (see Table 2). *B-rated senior debt* is equal to one if the senior debt associated with the loan facility is rated B1, B2, or B3 at loan origination. *Term structure* is calculated as the difference between the 5-year and 3-month T-bill rate in the month of deal origination. *Credit spread* is the spread of a ten-year Baa- and B-rated bond index over the ten-year T-bond rate, respectively. *Covenants indicator* is equal to one if a facility has financial covenants. *Pricing grid indicator* and *Revolving pricing grid indicator* are equal to one if the spread is adjustable based on accounting performance. *Size of syndicate* is the number of lenders involved in the lending syndicate. *Tranche maturity* and *Revolving maturity* represent the number of months until the Term Loan B and revolving facilities mature, respectively. *Log facility amount* and *Revolving facility amount* represent the dollar amount (\$ million) of the Term B and revolving loan facilities, respectively. *Log sales size* represents the sales of the borrowing firm, measured in U.S. \$ million. *Secured indicator* is an indicator equal to one if the loan facility is secured. We control for tranche-purpose fixed-effects (“corporate purposes,” “M&A,” “debt repayment,” etc.), year fixed-effects, and industry fixed-effects (two-digit SIC code) in all specifications and credit-rating fixed-effects in column 1. We restrict our sample to include syndicated loan facilities originated in the United States and exclude firms with SIC codes between 6000 and 6500. We also restrict our analysis to loan facilities with senior debt rated Ba1, Ba2, Ba3, B1, B2, and B3. We identify and remove any second-lien or mezzanine facilities from the sample. Standard errors are clustered by year and by borrowing firm. The symbols “*”, “”, and “*” indicate significance at the 1%, 5%, and 10% levels, respectively.

Dependent var.: Within deal spread	Term Loan B spread vs. Revolver spread in the same deal (1)	Term Loan B spread vs. Revolver spread in the same deal (2)	Term Loan B spread vs. Revolver spread in the same deal (3)
Securitization-active bank*B-rated sr. debt			–14.171* (1.90)
Securitization-active bank	–6.887** (2.55)	–6.798** (2.29)	2.971 (0.64)
B-rated sr. debt		–8.014 (1.50)	2.373 (0.47)
Term structure	3.823 (0.68)	4.369 (0.79)	4.406 (0.80)
Credit spread	13.278** (2.08)	12.087* (1.79)	12.084* (1.76)
Covenants indicator	4.353 (0.90)	5.940 (0.99)	4.587 (0.93)
Pricing grid indicator	–6.076** (2.10)	–6.106** (2.06)	–5.961** (2.02)
Revolving pricing grid indicator	–1.180 (0.34)	–3.138 (0.75)	–2.624 (0.65)
Size of syndicate	–0.028 (0.14)	–0.030 (0.15)	–0.034 (0.17)
Tranche maturity	–0.115 (0.75)	–0.190 (1.14)	–0.174 (1.10)
Revolving maturity	0.254 (1.46)	0.288 (1.56)	0.282 (1.54)
Log facility amount	–9.286*** (2.92)	–10.280*** (2.93)	–10.292*** (2.94)
Revolving facility amount	7.627*** (3.69)	8.770*** (4.46)	8.763*** (4.50)
Log sales size	–1.771 (1.51)	–1.300 (1.18)	–1.326 (1.20)
Secured indicator	–2.738 (0.39)	–0.990 (0.13)	–0.649 (0.09)
Constant	19.993 (0.38)	41.140 (1.08)	27.786 (0.76)
Moody's sr. debt fixed effects	Yes	No	No
Tranche-purpose fixed effects	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes
Standard errors clustered by:			
Borrower	Yes	Yes	Yes
Year	Yes	Yes	Yes
Adjusted R ²	0.130	0.111	0.113
Observations	1150	1150	1150

Table 8: Within-deal differences in loan spreads: Term Bs versus revolvers.

8 Result map

- Securitized facilities are cheaper.** Table 3 shows that securitized loans have lower spreads, with an estimated discount of roughly 17 basis points in the key specification.
- Term Loan B facilities are securitized more often.** Table 4 confirms that Term Loan B structure predicts securitization.
- Securitization-active banks matter.** Table 4 and Table 5 show that banks active in CLO underwriting are associated with both securitization likelihood and pricing discounts.
- B-rated loans matter.** B-rated borrowers are more likely to produce CLO-attractive collateral, and pricing discounts are concentrated in this group.
- Discount is specific to securitization-friendly Term Bs.** Table 6 shows that the discount is not replicated in Term Loan As or revolvers.
- Within-deal evidence supports the mechanism.** Tables 7 and 8 show that Term Loan B discounts persist relative to other facilities in the same deal.
- CLO ownership data support the rating mechanism.** Table 9 shows heavier CLO ownership in lower-rated Term Loan Bs.

Table 9

Differences in CLO ownership by credit rating.

This table reports data on CLO ownership of securitized Term Loan B facilities. A portion of the same loan facility can be owned by multiple CLOs. Panel A reports the sum total of CLO ownership as a percent of the total loan facility, sorted by Moody's senior debt credit rating at the time of loan origination. Panel B reports the average CLO ownership percentage in a given facility across all CLOs by credit rating. The upper portion of Panel C calculates the difference in total CLO ownership percentages between facilities with B-rated senior debt and facilities with Ba-rated senior debt. The lower portion of Panel C calculates the difference in total CLO ownership percentages between facilities with B2- and B3-rated senior debt compared to Ba1- and Ba2-rated senior debt. The upper portion of Panel D calculates the difference in average ownership percentages between facilities with B-rated senior debt compared to facilities with Ba-rated senior debt. The lower portion of Panel D calculates the difference in average ownership percentages between facilities with B2- and B3-rated senior debt compared to facilities with Ba1- and Ba2-rated senior debt. The ownership percentages reported in Moody's EMS database are as of the first quarter 2000.

Panel A: Total CLO ownership % of Term Loan Bs						Panel B: Average CLO ownership % of Term Loan Bs					
Moody's senior debt rating	N	Mean	Std. dev.	Avg. fac. amt. (\$M)		Moody's senior debt rating	N	Mean	Std. dev.	Avg. fac. amt. (\$ M)	
Ba1	10	3.08%	6.98%	744.4		Ba1	10	0.29%	0.22%	744.4	
Ba2	24	9.94%	11.88%	693.5		Ba2	25	0.53%	0.73%	717.2	
Ba3	52	17.34%	23.01%	667.2		Ba3	53	0.63%	1.02%	654.9	
B1	93	14.21%	20.74%	409.1		B1	95	0.95%	2.28%	402.1	
B2	61	15.40%	20.35%	569.1		B2	63	0.80%	0.98%	563.4	
B3	28	15.83%	26.93%	766.4		B3	28	0.50%	0.48%	766.4	
Panel C: Differences in total CLO ownership % of Term Loan Bs						Panel D: Differences in average CLO ownership % of Term Loan Bs					
B1, B2, B3 rated	Ba1, Ba2, Ba3 rated	Diff.	t-Stat.			B1, B2, B3 rated	Ba1, Ba2, Ba3 rated	Diff.	t-Stat.		
Mean	14.89%	13.62%	1.27%	(0.48)		Mean	0.85%	0.56%	0.29%	(1.84)	
N	182	86				N	186	88			
B2, B3 rated	Ba1, Ba2 rated	Diff.	t-Stat.			B2, B3 rated	Ba1, Ba2 rated	Diff.	t-Stat.		
Mean	15.59%	7.92%	7.67%	(2.52)		Mean	0.71%	0.46%	0.25%	(1.76)	
N	89	34				N	91	35			

Table 9: Differences in CLO ownership by credit rating.

9 Short summary

- The paper studies whether CLO securitization lowered the cost of corporate bank loans.
- It uses Dealscan loan facilities and Moody's CLO collateral data.
- The main outcome is the all-in-drawn spread over LIBOR.
- Securitized loans have spreads roughly 17 basis points lower than comparable non-securitized loans.
- Term Loan B facilities, B-rated debt, and securitization-active banks are the core securitization-friendly characteristics.
- These characteristics predict actual securitization in CLO collateral data.
- B-rated Term Loan B facilities originated by securitization-active banks receive about a 20.9 basis point discount.
- Triple-interaction tests show the discount is specific to Term Loan Bs, not Term Loan As or revolvers.
- Within-loan comparisons show Term Loan Bs are relatively cheaper than Term Loan As or revolvers in the same deal when CLO demand should be strongest.
- CLO ownership is higher in lower-rated Term Loan B facilities, consistent with the rating-based pricing results.
- The evidence supports the view that securitization lowered the cost of capital for firms issuing securitization-friendly corporate loans during the CLO boom.

10 Conclusion

10.1 Core conclusion

Nadauld and Weisbach show that securitization affected the primary market for corporate debt. Corporate loan facilities that were attractive as CLO collateral were issued at lower spreads, especially B-rated Term Loan B facilities originated by securitization-active banks.

10.2 Economic intuition

CLOs created demand for a particular kind of collateral: senior, bullet-like, lower-rated loans that could be pooled and tranching. Banks that were active in CLO underwriting had strong incentives to originate such collateral. To attract borrowers and generate collateral supply, these banks could offer lower loan spreads. The discount therefore reflects secondary-market demand feeding back into primary-market credit supply.

10.3 Why the evidence is persuasive

The paper does not rely on one comparison. It first shows a direct spread discount for securitized loans, then verifies which loans are actually securitized, then shows that the discount is concentrated in the securitization-friendly intersection of facility type, bank type, and rating. Finally, it compares facilities inside the same loan deal, reducing borrower-risk confounds.

10.4 What the paper adds to the securitization literature

The paper expands the securitization debate beyond mortgages and demonstrates that CLOs can influence corporate borrowing costs. It also shows that securitization affects loan origination, not merely the allocation of risk after loans are created.

10.5 Policy implications

If securitization lowers credit costs, it can improve financing access for firms. But it also means that structured-finance demand can reshape bank lending incentives. During booms, banks may originate assets because they are useful for securitization rather than because they are optimal to hold on balance sheet. This creates potential benefits in credit supply and potential risks in underwriting incentives.

10.6 Limitations

The paper cannot observe every securitization event because the Moody's collateral database is a snapshot. The empirical design is observational, and securitization-active banks could differ from other banks in unobserved ways. The within-loan tests mitigate but do not completely eliminate such concerns.

10.7 Takeaway

The central lesson is that securitization can reduce the cost of corporate debt by creating collateral demand. The effect is economically meaningful and concentrated exactly where CLOs' collateral preferences predict it should appear.